

Meshless generalized finite difference method with a domain-selection method for solving degenerate boundary problems

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ABSTRACT

This paper is dedicated to presenting a combined *Generalized Finite Difference Method* (GFDM) and *Domain-Selection Method* (DSM) approach for solving degenerate boundary problems which are usually encountered in engineering simulations. Compared with traditional *Domain-Decomposition Method* (DDM), the most critical advantage of the present GFDM-DSM is that it is entirely free from matching continuous boundary conditions for different parts of the fluid domain of interest, which is sometimes hard to perform for specific situations. Instead, the only prerequisite of the present GFDM-DSM is to identify which sub-domain each computational node belongs to, and then a linear algebraic system can be assembled according to the pattern of domain-selection. Four benchmarks are performed to validate the convergence and accuracy of the present GFDM-DSM. It is shown that the present GFDM-DSM holds satisfactory accuracy for solving degenerate boundary problems. In particular, the present DSM is a flexible strategy since it can be also combined with other localized domain-type meshless methods in practical applications.

1. Introduction

Degenerate boundary problems, also known as singular boundary problems, are very common in engineering. The main characteristic of degenerate boundary problems is the existence of flow discontinuity in a fluid domain due to the presence of an obstacle (also referred to as a degenerate boundary). For instance, when groundwater passes a sheet pile, the velocity field is inconsistent at different sides of the sheet pile. Such a phenomenon makes a numerically complex situation for many *Localized Domain-type Meshless Methods* (LDTMMs) (see e.g. [1–7]). More specifically, when a flow is separated by an obstacle, fluids on different sides of the obstacle cannot directly interact with each other from a physical point of view. This situation can be conveniently addressed in conventional mesh-based methods via enforcing proper boundary conditions at the degenerate boundary, while this becomes a challenging problem in many LDTMMs. The main reason is that in most LDTMMs a field quantity at a specific node is interpolated from its neighboring nodes via evaluating a localized weighting function. When searching the neighbors of a targeted node, it is possible to occur that two nodes located on different sides (physically separated) of a fluid domain are identified as an interacting node pair. Nevertheless, these

two nodes indeed cannot interact with each other from a physical point of view and hence special techniques are required to deal with such a situation.

At present, one popular strategy to tackle the aforementioned problem within the LDTMMs framework is to perform the technique so-called *Domain-Decomposition Method* (DDM). The main idea of DDM is to divide the whole computational domain into different parts and then a linear algebraic system can be yielded by exploiting the *continuous boundary conditions* at the interface of different sub-domains. For instance, Chen et al. [8] succeeded in modeling similar degenerate boundary problems by using the *Method of Fundamental Solutions* (MFS) in conjunction with DDM. Another instance corresponds to the works by Young et al. [9] as well as Tsai and Young [10] where MFS was combined with DDM to handle degenerate boundary problems in seepage flows with sheet piles, and water wave diffraction problems by thin porous breakwaters, respectively. In addition, several researchers also combined *Singular Boundary Method* (SBM) with DDM for the simulations of heat conduction problems in layered materials [11,12], and elastic mechanics problems [13].

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Despite the fact that DDM has been successfully combined with different meshless methods for solving degenerate boundary problems, it seems that combining DDM with *Generalized Finite Difference Method* (GFDM) has received limited attention in the past years. In fact, GFDM is a relatively young approach that was first proposed by Benito et al. in 2001 [14]. Since it was proposed, GFDM has achieved prosperous developments in the meshless method research community due to its capability for solving several challenging engineering problems. At present, GFDM has been applied to extensive engineering problems including coastal and ocean engineering [15–21], heat conduction engineering [22,23], material engineering [24,25], etc.

In terms of works related to GFDM-DDM, for example, in 2018 Wang et al. [24] have combined GFDM with DDM to analyze stresses in multi-layered elastic materials. Another example corresponds to the work by Huang et al. in 2023 [26] where GFDM has been combined with DDM to investigate caisson resonance problems occurring in coastal and ocean engineering.

Although the strategy of GFDM-DDM has shown great accuracy and robust performance in practical applications, it is not trivial to be implemented in some problems where continuous boundary conditions at the interface of different sub-domains are hard to be obtained. This paper aims at presenting a novel scheme called *Domain-Selection Method* (DSM) to solve the degenerate boundary problems within the GFDM framework. Compared with the previous GFDM-DDM [24,26], the main advantage of the present GFDM-DSM is that the novel scheme is entirely free from matching continuous boundary conditions during the assembling of the linear algebraic system. Instead, the only prerequisite of the present GFDM-DSM is to classify and mark computational nodes prior to computation, and then the linear algebraic system can be arranged according to the pattern of domain-selection to obtain final numerical results.

The rest of this article is organized as follows. In Section 2, governing equations and boundary conditions of the problem of interest are briefly introduced. In particular, the principle of the present DSM is described in detail within the GFDM framework. Furthermore, in Section 3 four validations are performed to validate the convergence and accuracy of the present GFDM-DSM. Finally, concluding remarks and future prospects are given in the last part of the paper.

2. Numerical model

2.1. Governing equation

The problem of concern in this paper is illustrated in Fig. 1a. We suppose that a degenerate boundary lies in a fluid domain Ω with an arbitrary border $\partial\Omega$. The fluid is assumed to be incompressible, inviscid and irrotational, meaning that the flow satisfies the 2D Laplace equation, which yields

$$\begin{cases} \nabla^2 \phi(x, y) = 0, & (x, y) \in \Omega, \\ \phi(x, y) = D(x, y), & (x, y) \in \partial\Omega^D, \\ \frac{\partial \phi(x, y)}{\partial \mathbf{n}} = \mathcal{N}(x, y), & (x, y) \in \partial\Omega^N, \end{cases} \quad (1)$$

where $\phi(x, y)$ denotes the potential function with x and y being the coordinate of the fluid domain of interest. $\mathbf{n}$ represents the unit outward normal vector of boundaries. $\partial\Omega^D$ and $\partial\Omega^N$ denote Dirichlet and Neumann boundaries, respectively. $D(x, y)$ and $\mathcal{N}(x, y)$ represent the general formulae satisfying the Dirichlet and Neumann boundary conditions, respectively.

2.2. Generalized finite difference method

In the previous subsection, the governing equation and boundary conditions are briefly introduced, while in this subsection the fundamental concepts of GFDM are described in detail. The main idea of

GFDM is to transform continua into discretized nodes for the whole computational region to obtain a linear algebraic system. As shown in Fig. 1b, considering the i th node of interest and its n_s nearest neighbors within the support domain, the Taylor series expansion can be adopted to derive the approximation of an arbitrary physical quantity of the i th node, which yields

$$\begin{aligned} \phi_{i,j} = & \phi_i + x_{i,j} \left. \frac{\partial \phi}{\partial x} \right|_i + y_{i,j} \left. \frac{\partial \phi}{\partial y} \right|_i + \frac{x_{i,j}^2}{2} \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i \\ & + \frac{y_{i,j}^2}{2} \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i + x_{i,j} y_{i,j} \left. \frac{\partial^2 \phi}{\partial x \partial y} \right|_i + \dots \quad (j = 1, \dots, n_s), \end{aligned} \quad (2)$$

where ϕ_i and $\phi_{i,j}$ represent the field quantities at the i th node and its neighboring nodes, respectively. $x_{i,j} = x_j^i - x_i$ and $y_{i,j} = y_j^i - y_i$ denote the distances between the i th node and the j th node in the support domain along with x and y direction, respectively.

Note that Eq. (2) can be truncated by the second-order derivative because the governing equation Eq. (1) is a second-order partial differential system. Subsequently, it is possible to define a function $B(\phi_{i,j})$ based on the *Moving-Least Squares* (MLS) theory [14], i.e.

$$B(\phi_{i,j}) = \sum_{i=1}^{n_s} [\mathcal{F}(\phi_{i,j}) W(x_{i,j}, y_{i,j})]^2, \quad (3)$$

where $\mathcal{F}(\phi_{i,j})$ is a residual function given by

$$\begin{aligned} \mathcal{F}(\phi_{i,j}) = & \phi_i - \phi_{i,j} + x_{i,j} \left. \frac{\partial \phi}{\partial x} \right|_i + y_{i,j} \left. \frac{\partial \phi}{\partial y} \right|_i \\ & + \frac{x_{i,j}^2}{2} \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i + \frac{y_{i,j}^2}{2} \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i + x_{i,j} y_{i,j} \left. \frac{\partial^2 \phi}{\partial x \partial y} \right|_i. \end{aligned} \quad (4)$$

In Eq. (3), $W(x_{i,j}, y_{i,j})$ stands for a weighting function and is adopted as the quadratic spline function in this paper. $W(x_{i,j}, y_{i,j})$ is written as

$$W(x_{i,j}, y_{i,j}) = \begin{cases} 1 - 6q^2 + 8q^3 - 3q^4 & d_{i,j} \leq d_i^{\max}, \\ 0 & d_{i,j} > d_i^{\max}, \end{cases} \quad (5)$$

where $d_{i,j} = \sqrt{x_{i,j}^2 + y_{i,j}^2}$ refers to the distance between the i th node and its neighbors, while $d_i^{\max}$ denotes the radius of the circular support domain. q is a distance coefficient defined as $q = d_{i,j}/d_i^{\max}$. For the sake of transforming the partial derivative system Eq. (4) into a linear algebraic system, we define a partial derivative matrix $\mathbf{D}_u$ as

$$\{\mathbf{D}_u\}_{5 \times 1} = \left\{ \left. \frac{\partial \phi}{\partial x} \right|_i, \left. \frac{\partial \phi}{\partial y} \right|_i, \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i, \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i, \left. \frac{\partial^2 \phi}{\partial x \partial y} \right|_i \right\}^T, \quad (6)$$

where the superscript T stands for the transpose of a matrix.

One can see that the norm of $B(\phi_{i,j})$ can be minimized with respect to the partial derivative $\mathbf{D}_u$ such that

$$\frac{\partial B(\phi_{i,j})}{\partial \mathbf{D}_u} = \mathbf{0}. \quad (7)$$

As a consequence, the following system is obtained, i.e.

$$\{\mathbf{A}\}_{5 \times 5} \{\mathbf{D}_u\}_{5 \times 1} = \{\mathbf{b}\}_{5 \times 1}, \quad (8)$$

where $\mathbf{A}$ is a symmetric matrix. $\mathbf{A}$ and $\mathbf{b}$ are respectively given as Eq. (9) and Eq. (10) (see Eq. 9 given in Box 1).

$$\{\mathbf{b}\}_{5 \times 1} = \begin{Bmatrix} -\phi_i \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j} + \sum_{j=1}^{n_s} \phi_{i,j} W_{i,j}^2 x_{i,j} \\ -\phi_i \sum_{j=1}^{n_s} W_{i,j}^2 y_{i,j} + \sum_{j=1}^{n_s} \phi_{i,j} W_{i,j}^2 y_{i,j} \\ -\phi_i \sum_{j=1}^{n_s} \frac{W_{i,j}^2 x_{i,j}^2}{2} + \sum_{j=1}^{n_s} \phi_{i,j} \frac{W_{i,j}^2 x_{i,j}^2}{2} \\ -\phi_i \sum_{j=1}^{n_s} \frac{W_{i,j}^2 y_{i,j}^2}{2} + \sum_{j=1}^{n_s} \phi_{i,j} \frac{W_{i,j}^2 y_{i,j}^2}{2} \\ -\phi_i \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j} y_{i,j} + \sum_{j=1}^{n_s} \phi_{i,j} W_{i,j}^2 x_{i,j} y_{i,j} \end{Bmatrix}, \quad (10)$$

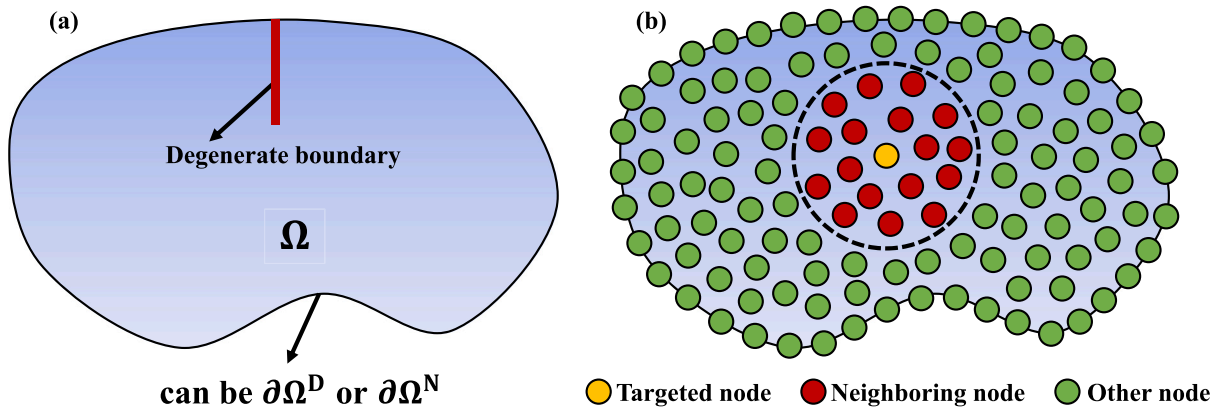


Fig. 1. Schematic diagram of the degenerate boundary problem (left) and the discretized computational domain (right).

$$\{\mathbf{A}\}_{5 \times 5} = \begin{Bmatrix} \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j}^2 & \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j} y_{i,j} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j}^3}{2} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j} y_{i,j}^2}{2} & \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j}^2 y_{i,j} \\ & \sum_{j=1}^{n_s} W_{i,j}^2 y_{i,j}^2 & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j}^2 y_{i,j}}{2} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{y_{i,j}^3}{2} & \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j} y_{i,j}^2 \\ & & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j}^4}{4} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j}^2 y_{i,j}^2}{4} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j}^3 y_{i,j}}{2} \\ & \text{Symmetry} & & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{y_{i,j}^4}{4} & \sum_{j=1}^{n_s} W_{i,j}^2 \frac{x_{i,j} y_{i,j}^3}{2} \\ & & & & \sum_{j=1}^{n_s} W_{i,j}^2 x_{i,j}^2 y_{i,j}^2 \end{Bmatrix}, \quad (9)$$

Box I.

where $W(x_{i,j}, y_{i,j})$ is simplified as $W_{i,j}$ for the sake of simplicity. With a clarity purpose, herein the matrix $\mathbf{b}$ can be re-written as

$$\mathbf{b} = \mathbf{BQ}, \quad (11)$$

where $\mathbf{Q} = \{\phi_i, \phi_{i,1}, \dots, \phi_{i,n_s}\}$ is the physical quantities at the i th node together with its neighboring nodes. Accordingly, $\mathbf{D_u}$ can be re-written as

$$\mathbf{D_u} = \begin{Bmatrix} \left. \frac{\partial \phi}{\partial x} \right|_i \\ \left. \frac{\partial \phi}{\partial y} \right|_i \\ \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i \\ \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i \\ \left. \frac{\partial^2 \phi}{\partial x \partial y} \right|_i \end{Bmatrix} = \mathbf{A}^{-1} \mathbf{b} = \mathbf{A}^{-1} \mathbf{BQ} = \mathbf{D} \begin{Bmatrix} \phi_i \\ \phi_{i,1} \\ \phi_{i,2} \\ \phi_{i,3} \\ \vdots \\ \phi_{i,n_s} \end{Bmatrix}_{(n_s+1) \times 1}. \quad (12)$$

For the convenience of readers, we recast the numerical scheme of the spatial derivative at the i th node as

$$\begin{cases} \left. \frac{\partial \phi}{\partial x} \right|_i = \sigma_{1,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{1,j+1}^i \phi_{i,j}, \\ \left. \frac{\partial \phi}{\partial y} \right|_i = \sigma_{2,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{2,j+1}^i \phi_{i,j}, \\ \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i = \sigma_{3,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{3,j+1}^i \phi_{i,j}, \\ \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i = \sigma_{4,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{4,j+1}^i \phi_{i,j}, \\ \left. \frac{\partial^2 \phi}{\partial x \partial y} \right|_i = \sigma_{5,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{5,j+1}^i \phi_{i,j}. \end{cases} \quad (13)$$

in which, $\{\sigma_{k,j}\}_{j=1}^{n_s+1}$ ($k=1, \dots, 5$) is the weighting coefficients, which can be numerically calculated using the aforementioned numerical procedures. According to Eq. (13), the governing equation Eq. (1) can be discretized as

$$\begin{cases} \left. \frac{\partial^2 \phi}{\partial x^2} \right|_i + \left. \frac{\partial^2 \phi}{\partial y^2} \right|_i = \sigma_{3,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{3,j+1}^i \phi_{i,j} + \sigma_{4,1}^i \phi_i \\ \quad + \sum_{j=1}^{n_s} \sigma_{4,j+1}^i \phi_{i,j} = 0, & i \in \Omega, \\ \phi_i = D_i(x, y), & i \in \partial\Omega^D, \\ \left. \frac{\partial \phi}{\partial x} \right|_i n_x + \left. \frac{\partial \phi}{\partial y} \right|_i n_y = \left(\sigma_{1,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{1,j+1}^i \phi_{i,j} \right) \cdot n_x \\ \quad + \left(\sigma_{2,1}^i \phi_i + \sum_{j=1}^{n_s} \sigma_{2,j+1}^i \phi_{i,j} \right) \cdot n_y = \mathcal{N}_i(x, y), & i \in \partial\Omega^N. \end{cases} \quad (14)$$

Finally, with a convenience purpose, we transform Eq. (14) into a matrix form as

$$\mathbf{C}\phi = \mathbf{K}, \quad (15)$$

where $\mathbf{C}$ is a coefficient matrix and the $\mathbf{K}$ the matrix of the inhomogeneous term and boundary term.

2.3. Domain-decomposition method

In this subsection, we introduce the fundamental concepts of DDM. The main idea of DDM is to separate a fluid domain into different parts and then each part can construct a linear algebraic system according to its domain conditions. The most critical point for DDM is the continuous boundary conditions at the interface of different parts, which can be exploited to combine all sub-domains to build a new linear algebraic system for the split domain.

For the sake of simplicity and clarity, we assume that the computational domain shown in Fig. 1a is divided into two parts (i.e. Ω_1 and Ω_2) as depicted in Fig. 2a. After splitting, the continuous boundaries of

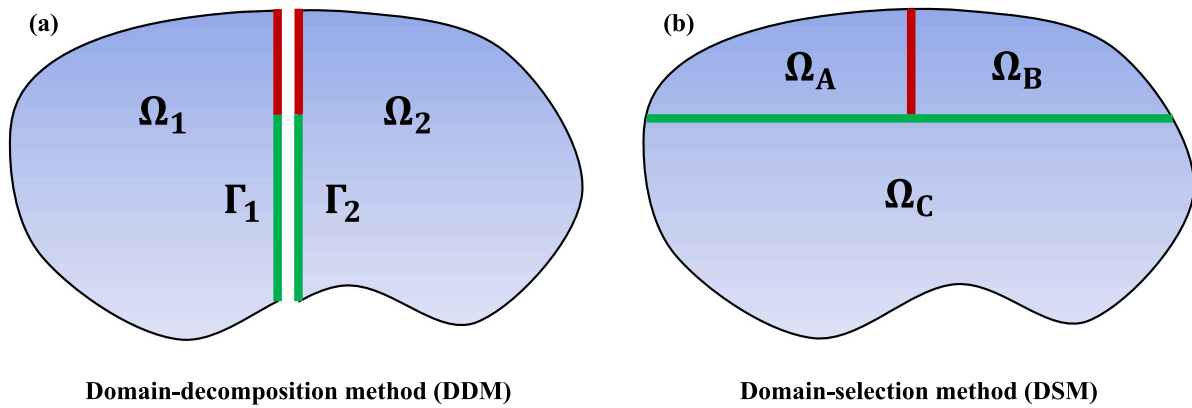


Fig. 2. Fundamental concepts of DDM and DSM.

Algorithm 1 Procedure of GFDM-DDM

- 1: Performing domain-decomposition and defining matching boundaries for the whole computational domain
- 2: Discretizing the whole computational domain into uniform nodes
- 3: Executing GFDM calculation
 - 3.1 Searching neighbors of each computational node for a sub-domain
 - 3.2 Computing the derivative matrix by utilizing Eqs. (3)~(13) for the sub-domain
 - 3.3 Repeating the procedure of 3.1 and 3.2 for each sub-domain
- 4: Building a linear algebraic system for each sub-domain by exploiting the governing equation Eq. (1)
- 5: Combining the algebraic system of each sub-domain by matching the continuous boundary condition Eq. (17)
- 6: Solving the combined linear algebraic system

Algorithm 2 Procedure of GFDM-DSM

- 1: Discretizing the whole computational domain into uniform nodes
- 2: Marking each computational node to determine which sub-domain they belong to
- 3: Executing GFDM calculation
 - 3.1 Searching neighbors of each computational node for every sub-domain
 - 3.2 Computing the derivative matrix by utilizing Eqs. (3)~(13) for the whole computational domain
- 4: Building a linear algebraic system by exploiting the governing equation Eq. (1)
- 5: Solving the linear algebraic system

Ω_1 and Ω_2 are symbolized as Γ_1 and Γ_2 , respectively. According to the principle of DDM, the discretized governing equations and continuous boundary conditions with respect to Fig. 1a can be written as

$$\begin{cases} C_1 \phi_1 = K_1, \\ C_2 \phi_2 = K_2, \end{cases} \quad (16)$$

and

$$\begin{cases} \phi_1|_{\Gamma_1} = \phi_2|_{\Gamma_2}, \\ \frac{\partial \phi_1}{\partial \mathbf{n}}|_{\Gamma_1} = \frac{\partial \phi_2}{\partial \mathbf{n}}|_{\Gamma_2}. \end{cases} \quad (17)$$

Combining Eqs. (16) and (17), we obtain

$$\begin{bmatrix} C_1|_{\Omega_1-\Gamma_1} & 0 \\ 0 & C_2|_{\Omega_2-\Gamma_2} \\ 1|_{\Gamma_1} & -1|_{\Gamma_2} \\ C_1|_{\Gamma_1} & -C_2|_{\Gamma_2} \end{bmatrix} \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} K_1 \\ K_2 \\ 0 \\ 0 \end{Bmatrix}, \quad (18)$$

where, for example, the subscript $\Omega_1 - \Gamma_1$ represents the coefficient matrix of those nodes in Ω_1 by excluding Γ_1 . The subscript Γ_1 denotes the coefficient matrix of those nodes on Γ_1 .

2.4. Domain-selection method

The main idea of the present DSM is illustrated in Fig. 2b. As one can see, in the present DSM the computational domain is separated

into different parts (herein three parts are adopted for simplicity). Prior to computation, all computational nodes are classified and marked according to the domain they belong to. During simulations, when looping the neighbors of a targeted node, only those nodes lying on specific sub-domains are considered.

Taking the scenario shown in Fig. 2b as an example, when a targeted node comes from Ω_A , only those nodes located on Ω_A and Ω_C can be regarded as neighbors during simulations. Herein, we adopt the symbol $A \leftrightarrow C$ to represent that two nodes respectively come from Ω_A and Ω_C can interact reciprocally. According to this definition, for all nodes located on Ω_A , only node pairs $A \leftrightarrow A$ and $A \leftrightarrow C$ are allowed. With the same principle, it is not difficult to see that for all nodes come from Ω_B , only node pairs $B \leftrightarrow B$ and $B \leftrightarrow C$ are accepted. Similarly, for all nodes comes from Ω_C , all possible interaction pairs, i.e. $C \leftrightarrow A$, $C \leftrightarrow B$, and $C \leftrightarrow C$, are permitted. As a consequence, the governing equations of the fluid domain shown in Fig. 2b can be written as

$$\begin{bmatrix} C_{AA} & 0 & C_{AC} \\ 0 & C_{BB} & C_{BC} \\ C_{CA} & C_{CB} & C_{CC} \end{bmatrix} \begin{Bmatrix} \phi_A \\ \phi_B \\ \phi_C \end{Bmatrix} = \begin{Bmatrix} K_A \\ K_B \\ K_C \end{Bmatrix}, \quad (19)$$

where, for example, C_{AC} represents the coefficient matrix of those nodes lying in Ω_A with all of their neighbors coming from Ω_C .

It should be underlined herein that, according to our experiences, making a sub-domain boundary perpendicular to a degenerate boundary would be a recommended strategy in practical applications to improve the numerical accuracy and stability of the present DSM.

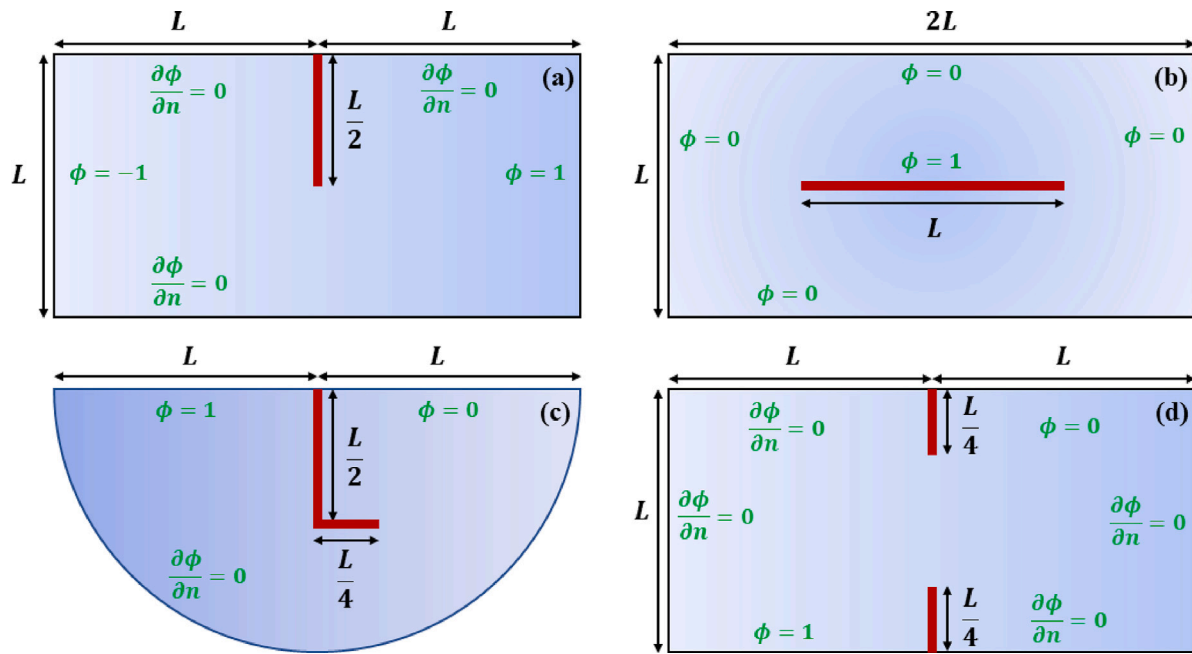


Fig. 3. Schematic diagrams of the four validation cases. (a) Groundwater flow with a sheet pile (see Section 3.1); (b) Electric potential field in a box caused by a strip conductor (see Section 3.2); (c) Seepage flow through a semi-circular soil stratum with a piece-wise sheet pile (see Section 3.3); (d) Groundwater flow with double sheet piles (see Section 3.4).

Table 1

Comparison of field distributions of $\partial\phi/\partial x$ along with y -axis at $x = 0$ between the present GFDM-DSM results and other reference results, where $\epsilon = (R_n - R_a)/R_a$ with ϵ , R_n and R_a denoting the relative error, numerical result and analytical solution, respectively.

y	Analytical [27]	BEM [28]	ϵ_{BEM}	MFS [9]	ϵ_{MFS}	GFDM-DDM	ϵ_{DDM}	GFDM-DSM	ϵ_{DSM}
0.01414	0.1096	0.1005	-8.30%	0.1123	2.46%	0.111	1.28%	0.112	2.19%
0.04243	0.1869	0.1803	-3.53%	0.1838	-1.66%	0.1873	0.21%	0.1891	1.18%
0.14142	0.3225	0.3117	-3.35%	0.3175	-1.55%	0.3223	-0.06%	0.323	0.16%
0.2609	0.4057	0.4119	1.53%	0.407	0.32%	0.4057	0.00%	0.4061	0.10%
0.28284	0.4159	0.4029	-3.13%	0.4194	0.84%	0.4159	0.00%	0.4172	0.31%
0.5	0.4614	0.4555	-1.28%	0.4728	2.47%	0.4626	0.26%	0.4617	0.07%

It is clear that, compared with DDM, the most critical advantage of the present DSM is that it is entirely exempted from matching the continuous boundaries connecting different parts of the fluid domain. The only prerequisite of the present DSM is to find out which sub-domain a computational node belongs to. During simulations, the linear algebraic system can be constructed based on the pattern of domain-selection. In other words, one can see that the merit of the sparse property of the linear algebraic system is retained in the present DSM without extra requirements, while the procedure of matching continuous boundary conditions in DDM is entirely exempted.

For the whole computing processes regarding DDM and DSM, ones can refer to Algorithms 1 and 2, respectively. In the next section, several tests will be performed to validate the feasibility and effectiveness of the present DSM.

3. Model validations

In this section, four benchmarks are performed to qualitatively and quantitatively validate the convergence and accuracy of the present GFDM-DSM. Among these benchmarks, the convergence performance of the present GFDM-DSM with respect to different resolutions and neighbor numbers is respectively highlighted in the first-two benchmarks, while in the last-two benchmarks, attention is particularly focused on the numerical accuracy compared with DDM. For the sake of clarity, the schematic diagrams of the four benchmarks are collected in Fig. 3. In the following discussions, $L = 1$ m is adopted for all cases for the sake of simplicity.

3.1. Validation No. 1: Groundwater flow with a sheet pile

The first validation case is related to groundwater passing a sheet pile. As shown in Fig. 3a, the computational region is a rectangle domain with a length $2L$ and a width L . A sheet pile is installed at the top side with a length $L/2$. At the beginning of the simulation, both the left and right sides are regarded as Dirichlet boundaries with an initial value $\phi = -1$ (left) and $\phi = 1$ (right), while the remaining boundaries are treated as Neumann boundaries.

First of all, the convergence performance of the present GFDM model with respect to different resolutions is validated. Fig. 4 shows field distributions of ϕ and $\partial\phi/\partial x$ for different resolutions. As one can see, the field distributions show almost identical characteristics with the increase in the number of total nodes, demonstrating that the present GFDM model holds satisfactory convergence performance. In particular, it can be observed that the field distributions of GFDM-DSM results and GFDM-DDM results are almost identical to each other, proving that the present GFDM-DSM can obtain the same accurate numerical results as GFDM-DDM.

Furthermore, in order to perform a quantitative validation of the accuracy of the present GFDM-DSM, in Table 1 the field distributions of $\partial\phi/\partial x$ along with y -axis at $x = 1$ are compared with analytical solutions [27], BEM solutions [28] and MFS solutions [9]. In addition, for the sake of clarity, in Fig. 5 the field distributions of $\partial\phi/\partial x$ along with y -axis at $x = 0$ are also plotted. It is clear from Table 1 and Fig. 5 that all GFDM-related numerical results are in better agreement with the analytical solutions than other numerical results. It is also found that GFDM-DSM almost holds the same accuracy as GFDM-DDM, proving that DSM, with a more easy-to-implement characteristic, retains the merit of DDM.

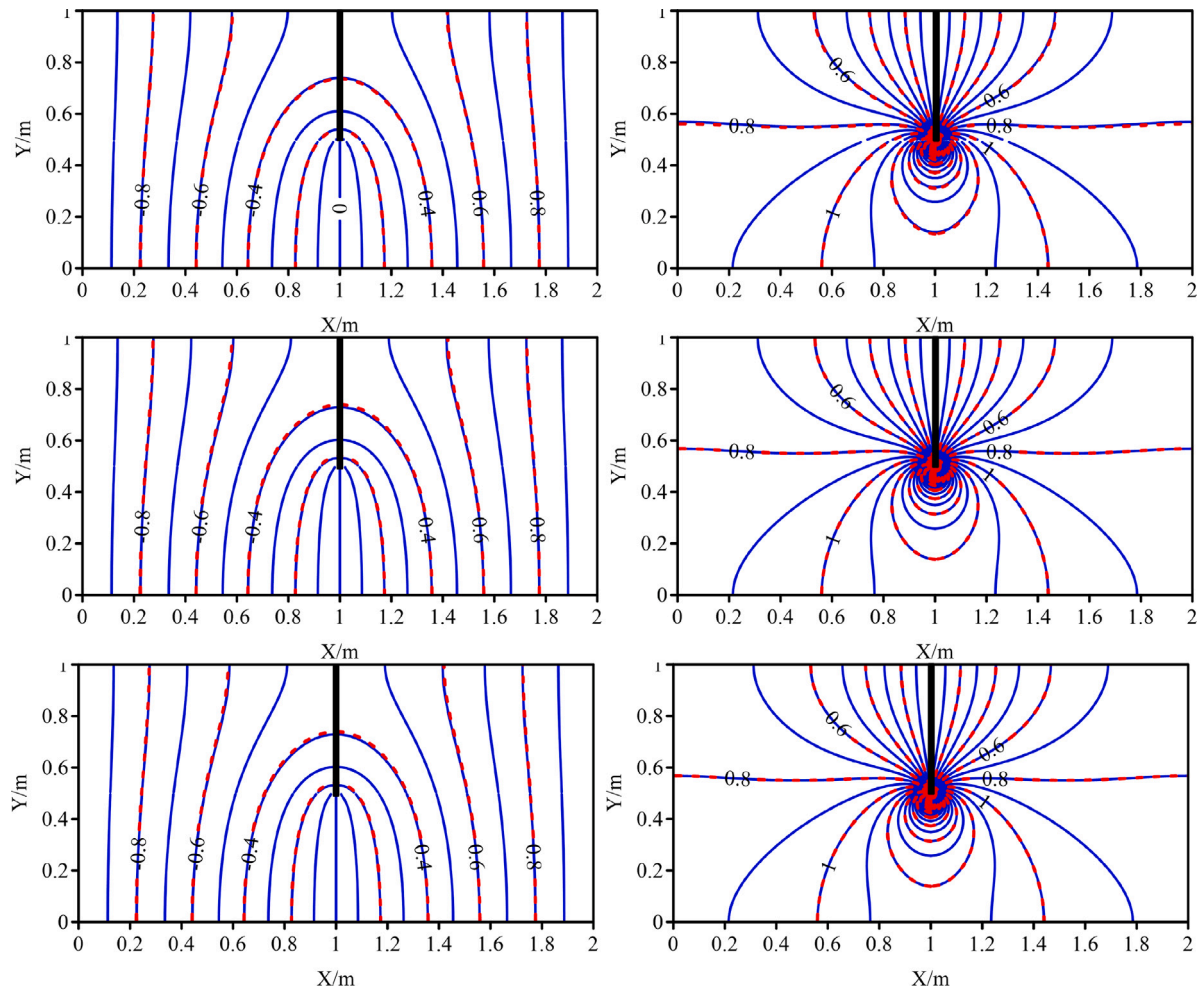


Fig. 4. Field distributions of ϕ (left) and $\partial\phi/\partial x$ (right) for different resolutions with $N_{\text{DSM}} = 20344$, $N_{\text{DDM}} = 20394$ (top), $N_{\text{DSM}} = 31683$, $N_{\text{DDM}} = 31746$ (middle) and $N_{\text{DSM}} = 80694$, $N_{\text{DDM}} = 80794$ (bottom). Note that hereinafter the dashed line and solid line denote DDM and DSM, respectively.

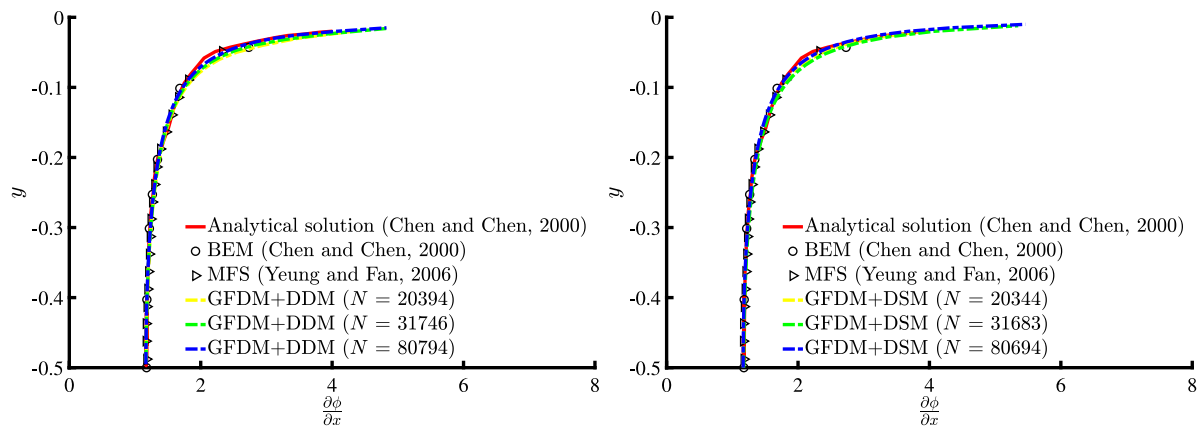


Fig. 5. Comparison between GFD-DM results (left), GFD-DSM results (right) and other reference results along with y -axis at $x = 0$.

3.2. Validation No. 2: Electric potential field in a box caused by a strip conductor

The second validation case is inspired by Ref. [29] where the electric potential field caused by a strip conductor enclosed in a shielded box containing a homogenous medium is investigated. In this benchmark, all sides and the strip conductor are treated as Dirichlet boundaries. As shown in Fig. 3b, the strip conductor with $\phi = 1$ is enclosed by

a rectangle box with a length $2L$ and a width L . $\phi = 0$ is enforced for all side boundaries. Besides, the number of total nodes for GFD-DSM and GFD-DM are $N_{\text{DSM}} = 20396$ and $N_{\text{DDM}} = 20494$, respectively.

We first test the effects of the number of neighboring nodes on the convergence performance of the present GFD-DSM. As shown in Fig. 6, the field distributions of ϕ and $\partial\phi/\partial y$ are almost identical with the increase of the number of neighboring nodes, demonstrating that $n_s = 15$ is sufficient to produce convergent numerical results for the present GFD-DSM. This n_s value is also adopted in the following.

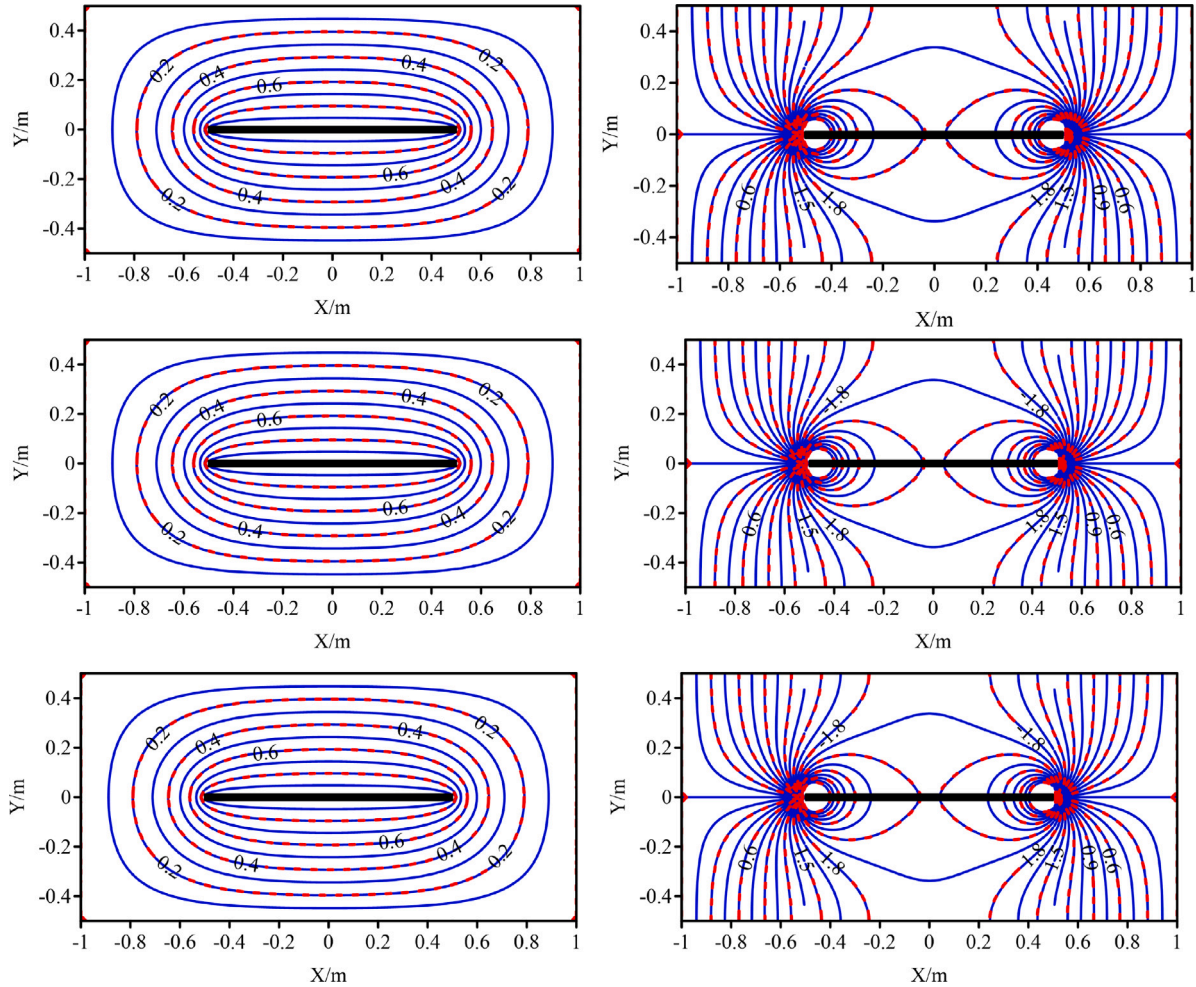


Fig. 6. Field distributions of ϕ (left) and $\partial\phi/\partial y$ (right) for different neighbor numbers with $n_s = 15$ (top), $n_s = 20$ (middle) and $n_s = 25$ (bottom).

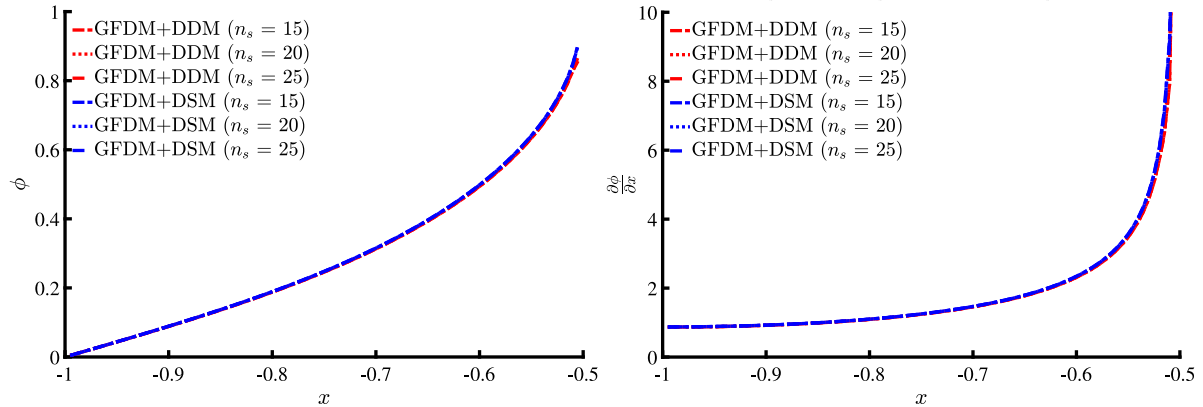


Fig. 7. Comparison between GFDM-DDM results and GFDM-DSM results for ϕ (left) and $\partial\phi/\partial x$ (right).

Fig. 7 illustrates the field distributions of ϕ and $\partial\phi/\partial x$ along with the x -axis at $y = 0$. Observing Figs. 6 and 7, it is evident that the numerical results obtained by GFDM-DSM and GFDM-DDM almost overlap each other, demonstrating that the present GFDM-DSM can provide the same accurate predictions as GFDM-DDM.

3.3. Validation No. 3: Seepage flow through a semi-circular soil stratum with a piece-wise sheet pile

The third benchmark, inspired by Ref. [9], aims at demonstrating the capability of the present GFDM-DSM in handling problems with an irregular domain. As shown in Fig. 3c, the radius of the semi-circular

stratum is adopted as L . A piece-wise sheet pile is attached at the top side of the semi-circular stratum, with a $L/2$ length vertical part and a $L/4$ length horizontal part. Note that in the original work [9], a steady hydraulic head is considered on the left side of the stratum, which causes a potential difference between the two sides of the stratum. Nevertheless, in this case, in order to take the hydraulic head into account in a simple manner, a Dirichlet boundary is directly enforced at the top-left side with an initial value of $\phi = 1$. The top-right side is also treated as a Dirichlet boundary with an initial value of $\phi = 0$, while the remaining boundary is treated as a Neumann type.

One can see that, compared to Ref. [9], the sheet pile is considered as a piece-wise type instead of a straight type that was originally

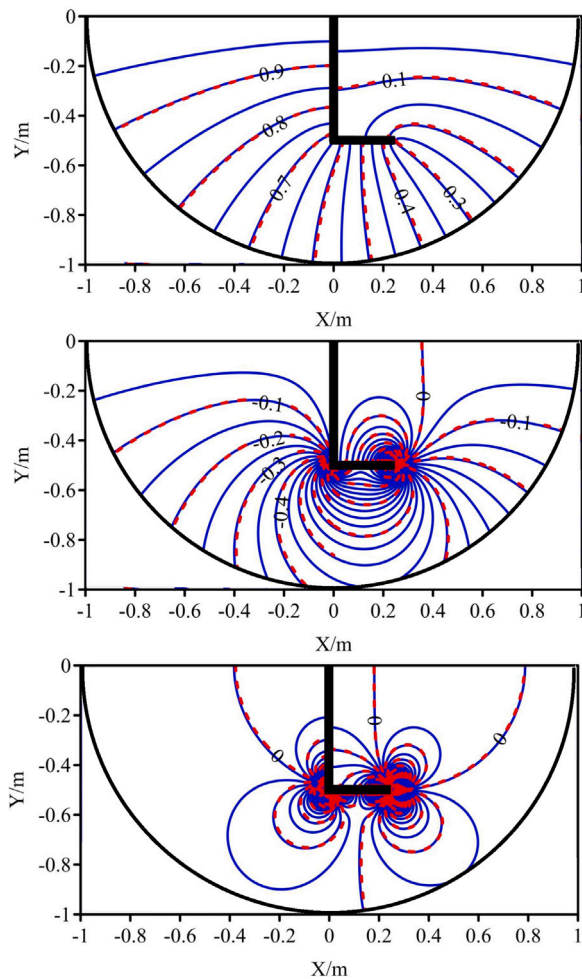


Fig. 8. Field distributions of ϕ (top), $\partial\phi/\partial x$ (middle) and $\partial^2\phi/\partial x^2$ (bottom).

adopted in Ref. [9]. This modification is to highlight the capability of the present GFDM-DSM in solving problems with a piece-wise boundary. Besides, the number of total nodes for GFDM-DSM and GFDM-DDM are $N_{\text{DSM}} = 63890$ and $N_{\text{DDM}} = 64070$, respectively.

In this benchmark, attention is particularly focused on the field distributions of ϕ , $\partial\phi/\partial x$ and $\partial^2\phi/\partial x^2$. Observing the top panel of Fig. 8, it can be found that the constant potential lines are perpendicular to the boundary of the soil stratum, which is caused by the fact that the flux at the wall boundary is zero. Such a phenomenon has been also captured in Ref. [9] via an MFS-DDM simulation. It is observed from the three panels that all numerical results almost overlap each other, demonstrating the accuracy of the present GFDM-DSM.

3.4. Validation No. 4: Groundwater flow with double sheet piles

The final benchmark is a modified version of the first one. As shown in Fig. 3d, the dimension of the computational domain is identical to Fig. 3a, while the sheet pile at the top side is shortened from $L/2$ to $L/4$ and a new sheet pile is installed at the bottom side with a length $L/4$. Both the left and right sides are treated as Neumann boundaries in this benchmark. In terms of the bottom and top sides, as one can see, the bottom-right and top-left sides are also treated as Neumann boundaries, while the bottom-left and top-right sides are regarded as Dirichlet boundaries with the initial values of $\phi = 1$ and $\phi = 0$, respectively. Besides, the number of total nodes for GFDM-DSM and GFDM-DDM are $N_{\text{DSM}} = 80695$ and $N_{\text{DDM}} = 80794$, respectively.

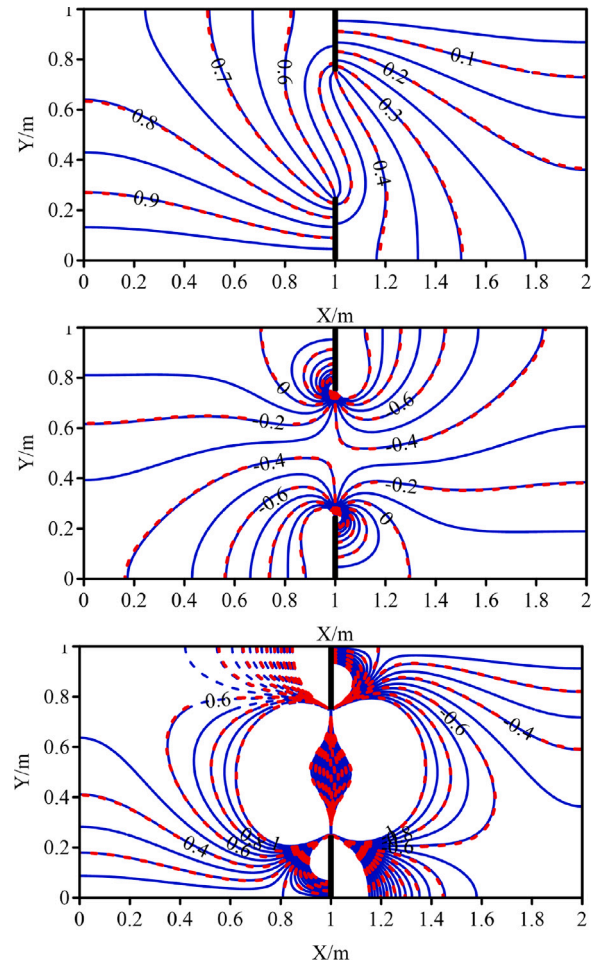


Fig. 9. Field distributions of ϕ (top), $\partial\phi/\partial y$ (middle) and $\partial^2\phi/\partial y^2$ (bottom).

Fig. 9 shows the field distributions of ϕ , $\partial\phi/\partial y$ and $\partial^2\phi/\partial y^2$ respectively from the top to bottom panel. It is proved that the numerical results obtained by GFDM-DSM and GFDM-DDM almost overlap each other, demonstrating again the accuracy of the present GFDM-DSM.

4. Concluding remarks and future prospects

This paper aims at presenting a combined GFDM and DSM approach (i.e. GFDM-DSM) for solving degenerate boundary problems. Compared with conventional GFDM-DDM, the most critical advantage of the proposed novel scheme is that the present GFDM-DSM is totally free from matching continuous boundary conditions at the interface of different parts of a fluid domain, which could be hard to be implemented in some complex problems. Instead, for the newly-proposed GFDM-DSM, the only prerequisite is to classify and mark all computational nodes coming from different sub-domains prior to computation. During simulations, the original GFDM linear algebraic system can be reformulated according to the pattern of domain-selection, which can be still efficiently solved using traditional linear algebra algorithms.

In order to validate the convergence and accuracy of the present GFDM-DSM, four benchmarks are performed with both qualitative and quantitative comparisons. It is demonstrated that the present GFDM-DSM, with a more easy-to-implement characteristic, can provide the same accurate numerical results as traditional GFDM-DDM.

It should be highlighted here that the proposed DSM strategy is not limited to GFDM but can be also implemented in other LDTMMs. In future studies, the present GFDM-DSM will be extended to fully 3D engineering problems with more complex case conditions.

CRediT authorship contribution statement

Ji Huang: Formal analysis, Methodology, Software, Resources, Validation, Investigation, Visualization, Writing – original draft. **Hong-Guan Lyu:** Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing – original draft. **Chia-Ming Fan:** Conceptualization, Methodology, Software, Resources, Data curation, Funding acquisition, Writing – review & editing. **Jiahn-Horng Chen:** Project administration, Resources, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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